AI-BASED DIGITAL ARCHIVE

Post-Demonstration Evaluation & Requirement Report

Scope: OCR Archive • Meeting Minutes • AI Search • Board Memo Summarization

Prepared for	Sonali Bank PLC
Purpose	Post-demonstration technical and business evaluation
Document Type	Solution Requirement / Evaluation Report
Status	For review and next-step assessment

1. Executive Summary

A demonstration session was conducted on the proposed AI-Based Digital Archive Solution. The solution demonstrated the use of Artificial Intelligence, Optical Character Recognition (OCR), Speech-to-Text, semantic search, and automated summarization to improve institutional document management, retrieval, meeting documentation, and Board-level information preparation.

Overall Assessment

The demonstrated capabilities are promising and potentially valuable for reducing manual effort, accelerating information retrieval, and improving institutional knowledge management. Before moving to the next phase, detailed technical sizing, AI architecture, security controls, licensing, integration requirements, and end-to-end business workflows are required.

2. Demonstrated Solution Scope

01 OCR-Based Digital Archive Digitize physical and scanned records, extract searchable text, classify documents, and preserve original files.	**02 Meeting Minutes - Speech-to-Text** Convert recorded or live meeting discussions into transcripts, decisions, action items, and draft minutes.
03 AI-Based Archive Search Search archived documents using natural language, keywords, metadata, and semantic retrieval with source references.	**04 Board Memo Summary** Analyze Board papers, generate concise summaries, highlight risks and decisions, and retrieve historical references.

3. Functional Evaluation

3.1 OCR-Based Digital Archive

The solution should support secure digitization and structured archival of institutional documents using OCR and document intelligence.

- Upload scanned documents, PDF files, images, letters, circulars, office orders, memorandums, Board papers, and other official records.
- Support Bangla and English OCR with measurable accuracy benchmarks.
- Extract document text and metadata such as title, reference number, date, department, subject, category, and keywords.
- Automatically classify, index, and tag documents while retaining the original source document unchanged.
- Enable full-text search, version history, role-based access, and complete audit trails.
- Provide human verification for low-confidence OCR and metadata extraction.

3.2 Meeting Minutes - Speech-to-Text

The solution should convert meeting audio or video into structured, reviewable meeting documentation.

- Accept recorded audio/video and support live transcription where technically feasible.
- Support Bangla and English speech recognition and speaker identification where possible.
- Generate transcript, discussion points, decisions, action items, responsible persons/divisions, and target dates.
- Prepare AI-assisted draft meeting minutes in an organizational template.
- Require authorized human review, correction, and approval before minutes become official.
- Archive audio/video, transcript, draft minutes, and approved minutes as linked records.

3.3 AI-Based Information Search

The solution should provide permission-aware intelligent search across archived institutional knowledge.

- Support natural-language questions, keywords, metadata filters, semantic search, and contextual retrieval.
- Allow search across OCR-extracted Bangla and English content.
- Return relevant documents, summaries, and answers with source references and document links.
- Retrieve related historical records and decisions based on context, not only exact keywords.
- Enforce document-level authorization before retrieval or AI response generation.
- Log search activity for audit and security review.

3.4 Board Meeting Memo Generation and Short Summary

The solution should assist management in reviewing Board memorandums and supporting documents while preserving human approval and accountability.

- Upload Board memos and supporting papers for OCR/document parsing and AI analysis.
- Generate concise summaries of lengthy Board papers.
- Identify subject, background, proposal, financial implications, risks, regulatory considerations, management recommendation, and required Board decision.
- Retrieve relevant previous Board decisions and historical references.
- Generate executive briefing notes for authorized users.
- Clearly distinguish AI-generated draft content from reviewed and approved official material.

4. System Requirements Required from Solution Provider

Mandatory Submission

The solution provider should provide separate sizing and architecture for Development, UAT, Production, and Disaster Recovery environments.

4.1 Hardware & Infrastructure Requirements

Area	Required Information
Compute	Application server, AI inference server, OCR processing server, database server, search/indexing server.
CPU / Memory	Recommended CPU cores, RAM, concurrency assumptions, and processing throughput.
GPU	Whether GPU is mandatory; recommended GPU model, VRAM, quantity, and workload sizing.
Storage	Primary document storage, object storage, database storage, index/vector storage, backup capacity, and annual growth.
Network	LAN/WAN bandwidth, latency expectations, load balancing, reverse proxy, firewall ports, and segmentation.
Availability	High Availability topology, clustering, failover, and Disaster Recovery design.
Environment Sizing	Development, UAT, Production, and DR sizing with capacity assumptions.

4.2 AI/ML Requirements

- OCR engine/model and Bangla/English accuracy benchmarks.
- Speech-to-Text model, supported languages, diarization capability, and accuracy metrics.
- Large Language Model (LLM), embedding model, reranking model, and vector database used by the solution.
- Model hosting architecture, model size, GPU requirement, and inference throughput.
- Open-source/commercial licensing and all recurring subscription dependencies.
- Internet or external cloud dependency, if any, including what data leaves the organization.

- On-premise/offline deployment capability for confidential institutional data.
- Model update, fine-tuning, governance, version control, and rollback mechanisms.

4.3 Software Platform Requirements

Component	Expected Detail
Operating System	Supported production operating systems and hardening guidance.
Database	Relational database, HA/replication approach, backup and restore.
Application Platform	Web/application framework, API layer, web server, background processing.
Search & AI	Search engine, vector database, AI runtime, model serving framework.
Storage	File/object storage and retention management.
Observability	Logging, monitoring, metrics, alerting, and audit monitoring.
Integration	REST/SOAP APIs, messaging, SSO/LDAP/AD, SIEM, backup, and enterprise integration protocols.
Licensing	All software, model, database, OS, middleware, and third-party license implications.

5. Security & Governance Requirements

- Role-Based Access Control (RBAC) and document-level authorization.
- Integration with Active Directory/LDAP/SSO where required.
- Multi-Factor Authentication for privileged or sensitive access.
- Encryption in transit using TLS 1.2 or higher and encryption of data at rest.
- Comprehensive audit logging for upload, view, search, download, update, approval, and deletion.
- Administrative activity monitoring and SIEM integration.
- Backup encryption, secure retention, recovery testing, and Disaster Recovery controls.
- Prevention of unauthorized AI access to restricted documents.
- Clear disclosure of any external/cloud AI processing and data transfer.
- Vulnerability assessment, penetration testing, patching, and secure configuration management.

Critical Control

AI responses must inherit the user's document permissions. Restricted documents must never be retrieved, embedded in context, summarized, or exposed to unauthorized users.

6. Required End-to-End Business Workflows

6.1 OCR-Based Digital Archive

Document Collection → Scan/Upload → File Validation → OCR Processing → Text & Metadata Extraction → Classification → Human Verification → Approval → Archive Storage → Search Indexing → Role-Based Access → Audit Logging

6.2 Meeting Minutes

Meeting Conducted → Audio/Video Capture → Speech-to-Text → Speaker Identification → Transcript → AI Summary → Decision/Action Extraction → Draft Minutes → Authorized Review → Correction → Approval → Final Minutes → Archive & Indexing

6.3 AI-Based Archive Search

User Login → Authorization → Question/Search → Permission Check → Keyword/Semantic Retrieval → Relevant Document

Selection → AI Context Processing → Response Generation → Source Citation → Original Document Access → Audit Log

6.4 Board Memo & Summary

Board Memo Upload → OCR/Parsing → AI Analysis → Metadata Extraction → Historical Search → Previous Decision Retrieval → Draft Summary → Risk/Financial/Regulatory Point Extraction → Management Review → Approval → Board Preparation → Final Archive

7. Integration, Performance & Capacity

7.1 Integration Requirements

- Active Directory / LDAP / SSO
- Email and notification services
- Existing Document Management / Digital Archive / ECM platforms
- Workflow or Board Management System
- API Gateway and enterprise integration layer
- SIEM, logging, backup, and Disaster Recovery platforms
- REST/SOAP APIs and supported integration standards

7.2 Performance & Capacity Information

Measure	Vendor to Specify
Concurrent users	Maximum supported active and concurrent users.
Document throughput	Documents processed per hour/day, including OCR and metadata extraction.
Audio processing	Speech-to-Text processing speed and maximum audio/video duration.
Search	Typical and peak search response time.
AI response	Typical AI response latency and throughput.
Scale	Maximum document count, index size, vector count, and storage volume.
Growth	Estimated annual storage/index growth and scaling approach.

8. AI Accuracy & Quality Validation

The provider should submit measurable accuracy and benchmark information for the following capabilities, and a Proof of Concept should preferably be performed using representative organizational documents and meeting samples.

- Bangla OCR
- English OCR
- Bangla Speech-to-Text
- English Speech-to-Text
- Document classification
- Metadata extraction
- AI summarization
- Semantic retrieval / search
- Question answering with source grounding

Recommended POC Principle

Evaluation should use real-world document quality, bilingual content, representative Board papers, and realistic audio conditions. Results should include accuracy, latency, failure cases, and human correction effort.

9. Information Required Before Next Phase

☐ Detailed Solution Architecture	☐ Hardware Sizing Document
☐ AI/ML Architecture and Model Details	☐ Software Component and License List
☐ Cybersecurity Architecture	☐ Deployment Architecture
☐ High Availability and DR Architecture	☐ Detailed Business Workflow for Each Module
☐ Integration Architecture	☐ API Documentation
☐ Data Flow Diagram	☐ User Access and Authorization Model
☐ AI Accuracy / Benchmark Report	☐ Bangla Language Capability Details
☐ Document Storage and Retention Architecture	☐ Implementation Methodology
☐ Project Timeline	☐ Training and Knowledge Transfer Plan
☐ Support and Maintenance Model	☐ Estimated Infrastructure Cost
☐ Estimated Software / License Cost	☐ Proof of Concept Proposal

10. Conclusion & Recommended Next Step

The AI-Based Digital Archive demonstration presented useful capabilities for institutional digitization, knowledge retrieval, meeting documentation, and Board information preparation. The solution may provide meaningful operational benefits, but technical feasibility, security, scalability, cost, AI accuracy, data governance, and business-process fit must be validated before implementation.

Recommended Next Step

Request the solution provider to submit the complete system requirements and detailed end-to-end business workflows described in this report, followed by a controlled Proof of Concept and technical/security evaluation before procurement or production planning.

11. Evaluation Status

Demonstration	Completed
Technical Requirements	Pending vendor submission
Business Workflow	Pending vendor submission
Proof of Concept	Recommended
Production Decision	Subject to technical, security, business and commercial evaluation